Band-1 Hadamard Order 5156 — Miyamoto (Double Paley-I seed H_1288); Miyamoto-only Bonus (NOT in SPK); Gap 2052 Still OPEN

H_644/H_1288/C_1290/H_5156 PASS; 5156 Miyamoto-only bonus (NOT in SPK); gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

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Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-15_Band1_Hadamard_4948_Miyamoto`; `2026-09-15_Band1_Hadamard_4868_Miyamoto` (Double Paley-I pattern).
Lab. `/workspace/experiments/band1/hadamard_5156_miyamoto/`.
Cutoff. 2026-09-15T18:40:37Z (~2:40PM ET).

Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 644, 1288 and 5156 only** (verified $\max|HH^\top - nI| = 0$), via Miyamoto with Double(Paley-I) seed H_{1288} (H_{644} = Paley-I($q = 643$); H_{1288} = Sylvester-double(H_{644})). Paley-II path for H_{1288} is **FAIL** ($q = 643 \equiv 3 \pmod 4$; needs $\equiv 1$). Paley-I direct for H_{1288} is **SKIP** ($q = 1287 = 3^2 \cdot 11 \cdot 13$ not prime). It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under Sylvester∪Paley∪Kronecker (SPK) closure. Order 5156 is a **Miyamoto-only bonus** (NOT in SPK): Paley-I FAIL ($q = 5155 = 5 \cdot 1031$), Paley-II FAIL ($q = 2577 = 3 \cdot 859$ not prime power), Kronecker FAIL (4×1289). `clay_closed = false`.

Abstract

Prior Band-1 ships constructed Miyamoto-eligible orders through 4948 on box. This increment ships **order 5156** ($5156 = 4 \cdot 1289$, 1289 prime $\equiv 1 \pmod 4$) via Miyamoto with a **Double(Paley-I) seed** H_{1288}. Symmetric Paley conference C_{1290} over prime 1289; $H_{5156} = \text{Miyamoto}(H_{1288}, C_{1290})$. Unit tests **59/59 PASS**. SPK next gap remains **2052** (OPEN). Order 5156 is **Miyamoto-only** (NOT in SPK). Seeds H_{644}/H_{1288} enter SPK via Double Paley-I. The **full Hadamard conjecture remains OPEN**.

Keywords. Hadamard matrix; order 5156; Miyamoto; Paley type I; Double Paley-I; Paley conference; constructive gap 2052; SPK; Band-1; Miyamoto-only bonus; non-claim.

1. Statement lock

Definition. A Hadamard matrix of order n is an $n \times n$ matrix H with entries in ± 1 satisfying $HH^\top = nI$.

Finite lemmas (this increment). 1. H_{644} constructed and verified (Paley type I, $q = 643$). 2. H_{1288} constructed and verified (Sylvester double of H_{644}). Paley-II path for H_{1288} **FAIL** ($q = 643 \equiv 3 \pmod 4$). Paley-I direct **SKIP** ($q = 1287 = 3^2 \cdot 11 \cdot 13$). 3. C_{1290} constructed and verified (Paley conference, $q = 1289$): $CC^\top = 1289I$, zero diagonal. 4. H_{5156} constructed and verified (Miyamoto). **Not** the Hadamard conjecture. 5. SPK: $5156 \notin$ SPK (Miyamoto-only bonus). Paley-I / Paley-II / Kronecker paths for 5156 **rejected**. Next constructive gap remains **2052** (OPEN). Verified H_{644}/H_{1288} enter SPK as Double Paley-I seeds. Prior 4948 remains in SPK via Paley-II ($q = 2473$); prior 1236 remains in SPK via Paley-II ($q = 617$).

2. Arithmetic and eligibility

Quantity	Value
Target order	$5156 = 4 \cdot 1289$
$q = 1289$	prime, $1289 \equiv 1 \pmod 4 \rightarrow$ Miyamoto-eligible
Needed H_{q-1}	H_{1288}
Needed conference	order $1290 = q + 1$
H_{1288} path (used)	Double(Paley-I): $H_{644} = \text{Paley-I}(q = 643)$; $H_{1288} = \text{Sylvester-double}(H_{644})$
H_{1288} Paley-II	**FAIL**: $2(q+1) = 1288 \Rightarrow q = 643 \equiv 3 \pmod 4$ (needs $\equiv 1$)
H_{1288} Paley-I direct	**SKIP**: $q = 1287 = 3^2 \cdot 11 \cdot 13$ not prime
Conference path	Symmetric Paley over prime 1289
Paley-I for 5156	**REJECTED**: $q' = 5155 = 5 \cdot 1031$ not prime
Paley-II for 5156	**REJECTED**: $q' = 2577 = 3 \cdot 859$ not prime power

Quantity	Value
Kronecker for 5156	**REJECTED**: only 4×1289; $1289 \not\equiv 0 \pmod 4$ (not admissible)

3. Construction results

Object	Method	$\max \lvert HH^\top - nI \rvert$ (or conf)	Wall (s)	Status
H_{644}	Paley type I, $q = 643$	0	0.3404	**PASS**
H_{1288}	Sylvester double of H_{644}	0	0.9463	**PASS**
H_{1288} Paley-II	$q = 643 \equiv 3 \pmod 4$	—	—	**FAIL** (needs $\equiv 1$)
H_{1288} Paley-I direct	$q = 1287 = 3^2 \cdot 11 \cdot 13$	—	—	**SKIP** (not prime)
C_{1290}	Paley conference, $q = 1289$	0	1.8095	**PASS**
H_{5156}	Miyamoto(H_{1288}, C_{1290})	0	63.5549	**PASS**

Primary pipeline wall (incl. unit tests + SPK refresh) ≈ 68.3443 s. Compressed ± 1 matrices saved as `H644_pm1.npz`, `H1288_pm1.npz`, `C1290_pm1.npz`, `H5156_pm1.npz` (int8).

4. SPK closure note

Seeds: all admissible ≤ 2000 ∪ on-box prior classical/SPK ships (through 4948/4804/4036/3908/… + **1020/1116/1228/1236** Paley-II + **1032/1092/1152** Paley-I + **524/1048/548/1096/564/1128/600/1200/6** Double Paley-I + **4804** Paley-II prime power 7^4 + **4948** Paley-II $q = 2473$; **exclude** Miyamoto-only 4916/4868/4612/4516/4468/4388/4372/4196/4132/4084/3716/3524/3284/3076/3044/2836/2692/2564) ∪ verified H_{644}/H_{1288} (Double Paley-I) ∪ Sylvester ∪ Paley I/II (incl. prime powers), closed under Kronecker (limit 5600). Order 5156 is **Miyamoto-only** (not folded into SPK).

Quantity	Value
Known count	1140
5156 in SPK	**False** (Miyamoto-only bonus)
1288 in SPK	**True** (Double Paley-I seed)
644 in SPK	**True** (Paley-I seed)

Quantity	Value
4948 in SPK	**True** (Paley-II $q = 2473$)
1236 in SPK	**True** (Paley-II $q = 617$)
Next constructive gap	**2052** (OPEN)

5. Gap 2052 (method triage / non-finding ONLY)

Paley-I, Paley-II, Kronecker, and Miyamoto-with-Paley-conference paths for order 2052 all **FAIL** on-box arithmetic. This is a **non-finding**, not a counterexample. Gap **2052 remains OPEN**. Conjecture **OPEN**.

6. Non-claims

- No full Hadamard conjecture claim.
- `clay_closed = false` forever in this ship.
- GATE-04 collection **LOCKED**; no invented J.
- No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / finished-ToE claims.
- Order 5156 is a **Miyamoto-only bonus** (NOT in SPK).

7. Next optional

- q=1297 → n=5188 (Double Paley-I half=648 / SPK Paley-II q=2593)
- method-attack SPK gap **2052**
- Track B privacy / Object 1 meter await
- Skip 4996 unless H_312 becomes available (1248=4×312 Kronecker blocked)

8. Reproducibility

```
cd /workspace/experiments/band1/hadamard_5156_miyamoto
python3 verify_hadamard_5156.py
```

Artifacts: `results.json`, `literature_ledger.json`, `SUMMARY.txt`, `*.npz`, this PAPER.